The Predictability of Crime in London using Graph Neural Networks

Theo Dal Pozzo Dos Santos - 700006758

M.Sci Computer Science Thesis 2024

Overview

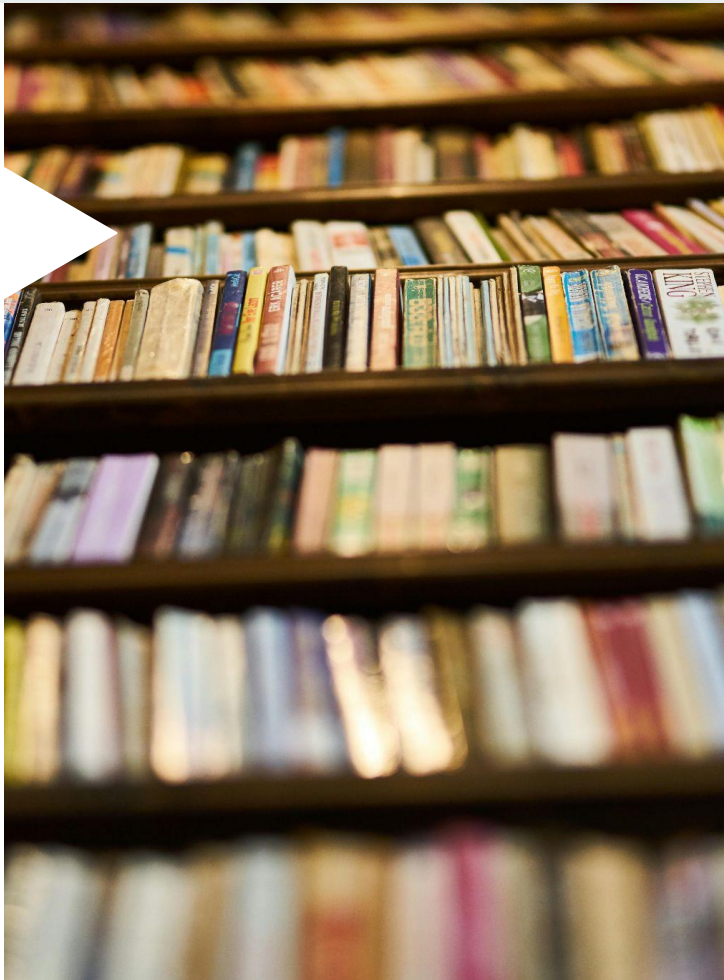
- London: 'Crime Capital of Europe'
- Persistent issues
- Need for data-driven approaches
- Project's novel approach

Project Objectives

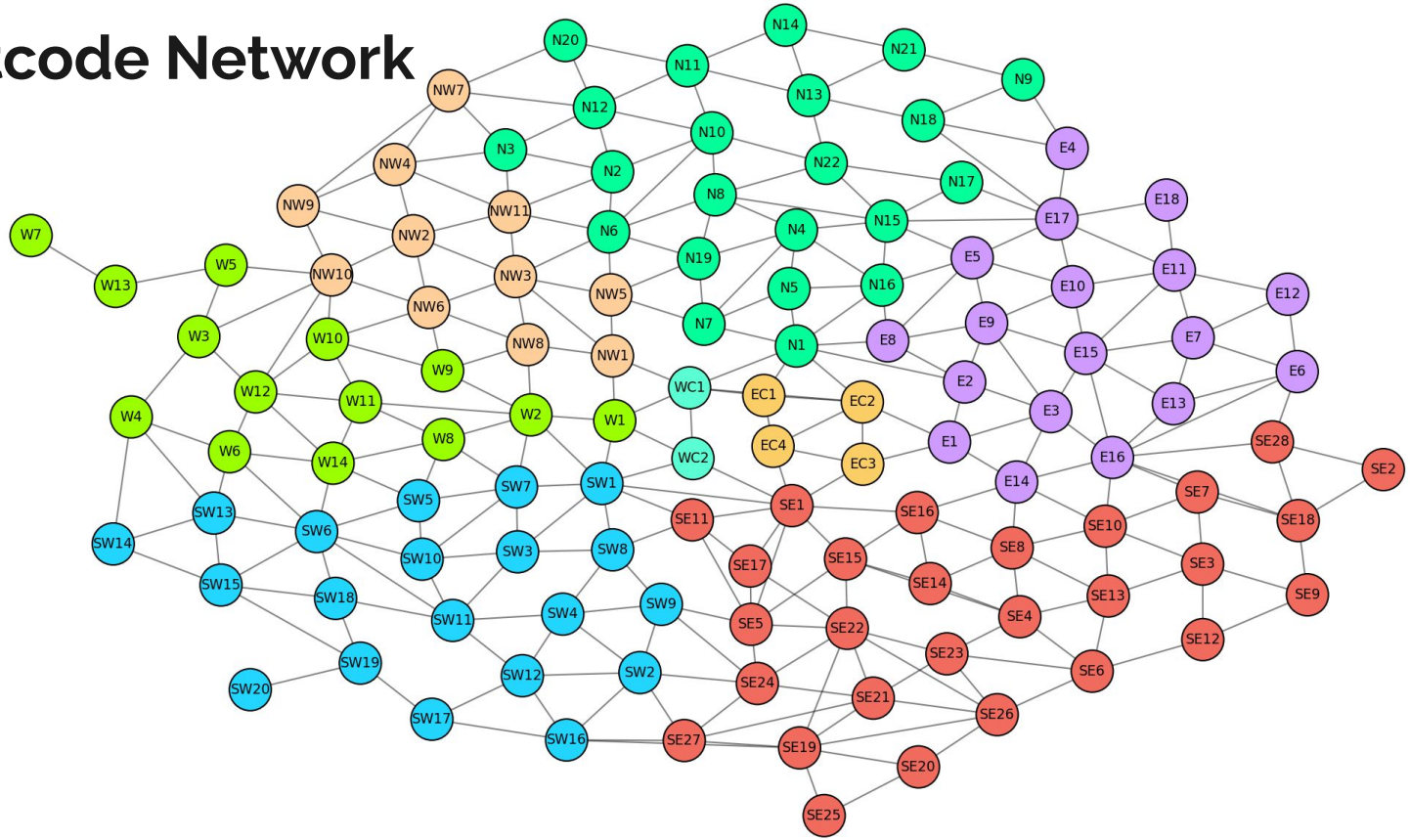
- Create the London Postcode Network (LPN)
- Calculate crime connectivity scores
- Develop GNN models for crime prediction
- Improve law enforcement resource allocation

Data Collection & Preprocessing

- Source: UK Police website
- 2019 crime data
- Approximately 500,000 crimes



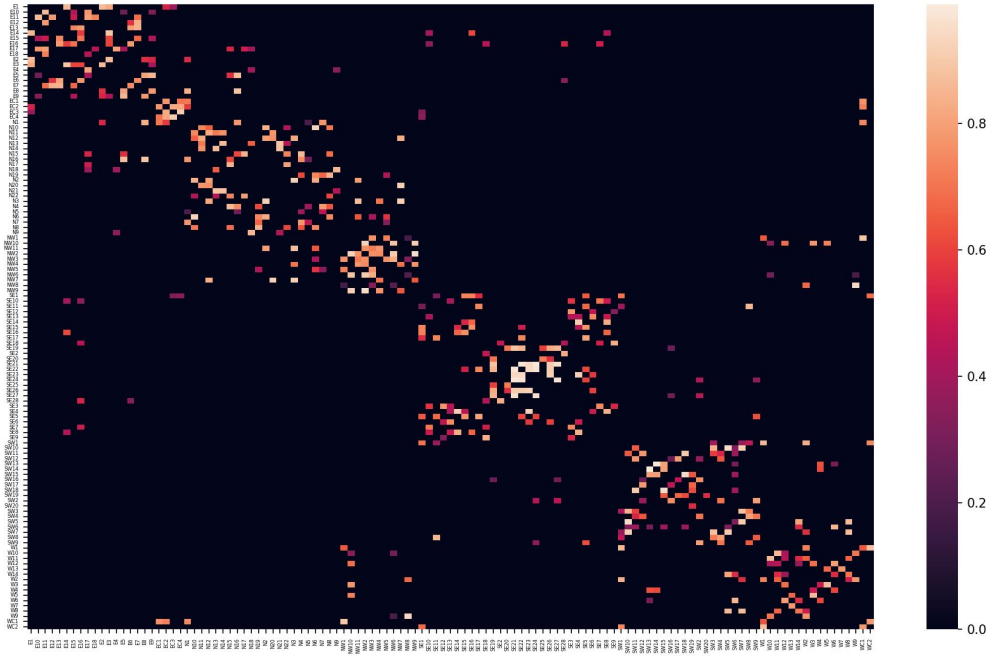
London Postcode Network



Postcode Crime Connectivity

Crime Type	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Anti-social behaviour	16	11	12	484	12	16	9	684	536	14	444	2
Bicycle theft	3	1	2	66	10	6	4	44	53	6	62	2
Burglary	2	6	1	103	6	10	3	91	72	2	74	1
Criminal damage and arson	6	3	4	67	2	3	0	50	46	4	48	1
Drugs	12	8	2	83	3	6	9	83	73	11	113	11
Other crime	0	0	1	10	1	1	1	11	12	2	13	0
Other theft	20	34	14	240	40	35	15	144	151	34	177	6
Possession of weapons	1	1	0	10	1	0	0	9	5	0	11	1
Public order	2	4	1	56	4	2	1	72	68	5	71	2
Robbery	4	1	1	60	2	5	1	37	33	3	44	0
Shoplifting	2	4	0	28	2	5	1	21	30	1	23	0
Theft from the person	14	11	2	85	18	15	3	73	55	25	68	1
Vehicle crime	4	3	1	89	3	5	1	86	103	1	100	3
Violence and sexual offences	4	13	6	255	16	26	4	258	269	21	262	7

Table 5.1: 2019 Crime Data for Postcode E1



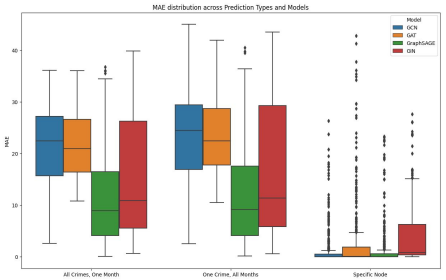
GNN Models Overview

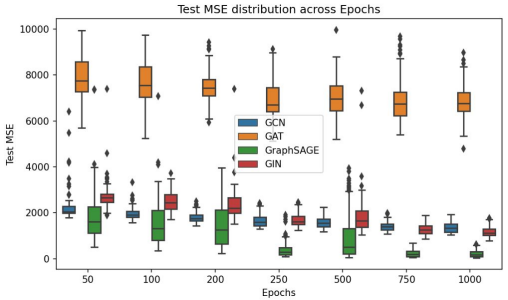
- GCN
- GAT
- GraphSAGE
- GIN

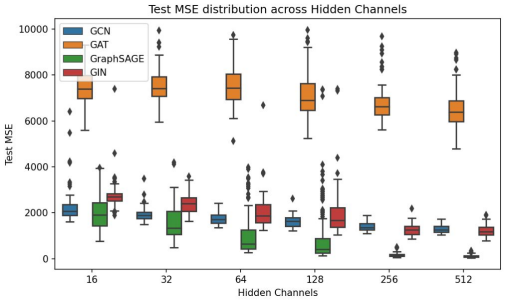
Predictive Tasks

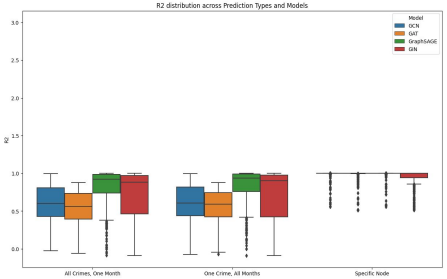
- One Crime Type Across All Months
- All Crime Types for One Month
- Specific Node Prediction

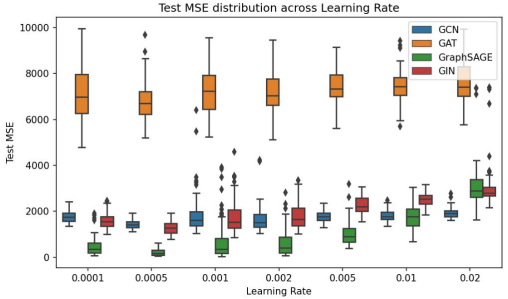
Results and Performance

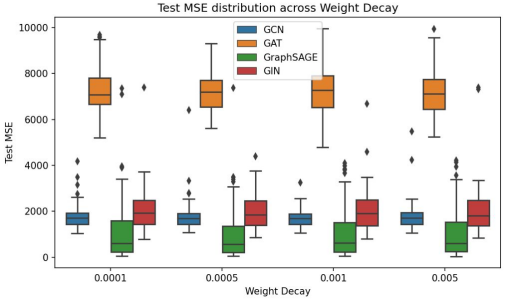












Key Findings

- GNNs effective for crime pattern prediction
- Edge connectivity improves accuracy
- GraphSAGE superior across tasks
- Spatial and temporal factors crucial

Implications for Crime Prevention

- Enhanced prediction accuracy
- Optimized resource allocation
- Targeted interventions
- Ethical considerations

Future Directions

- Advanced feature engineering
- Additional data sources
- Expand to other regions
- Real-time prediction capabilities

Closing Remarks

- GNNs effective for urban crime prediction
- Balancing technology and ethics
- Enhancing urban safety through data

Thank you

Link to code:

https://universityofexeteruk-my.sharepoint.com/:u:/g/personal/td442_exeter_ac_uk/EfmM5EbC9I1ApC7jsQc_JnsB4mxiZMXqUIlnBqcVedS_9A?e=IA6nkh

Link to video:

https://universityofexeteruk-my.sharepoint.com/:v:/g/personal/td442_exeter_ac_uk/ESWevXrNmwBBqu0SNjKtulwBFznxJw4yJyG_XD2gjguNIg?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=W8EqCX